

Operational Readiness for Object Detection

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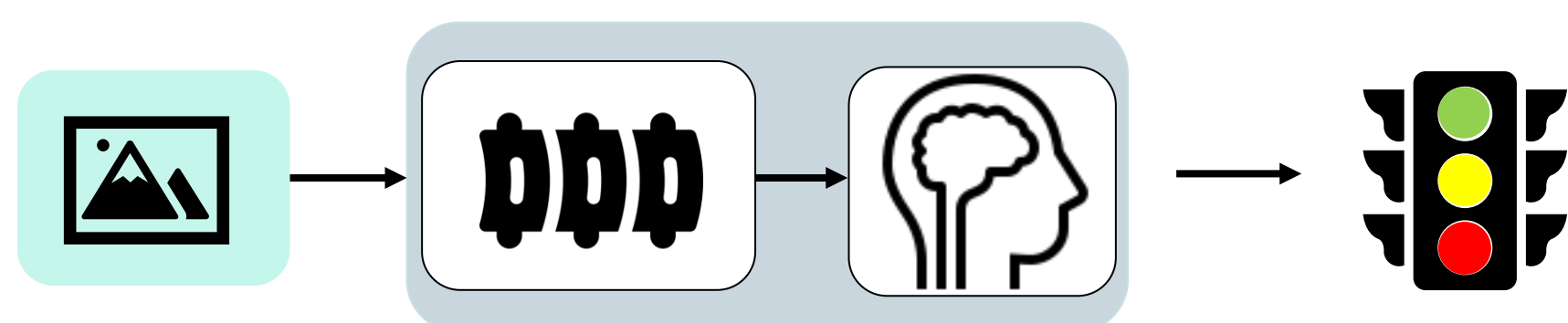
Motivation

“What is operational readiness?”



Operational readiness is modeled as a data-side gate:

Estimate whether an input image is drawn from the safe in-distribution.



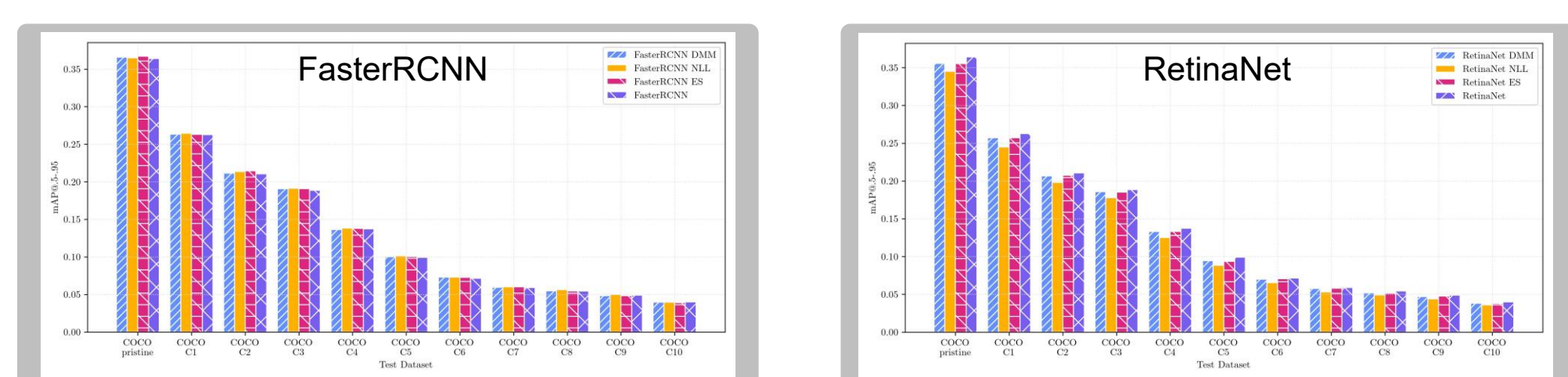
$$P_{OR}(\mathbf{o}|I, \mathcal{D}) = P(\mathbf{o}|I, \mathcal{D}) P_{OR}(I)$$

Key design requirements

- No unsafe labels
- Plug-in monitor
- Frozen detector

Dataset COCO-C (Corruption)

- Covariate-shifted version of COCO (Lin et al., 2014; [3])
 - Size: #img: 123k; #labels: 896k; 80 categories
- 18 corruptions from 7 groups (Hendrycks et al., 2019; [2])
 - Brightness change, blur, spatial distortions, noise, color distortion, sharpness/contrast, compression
- Extended severity levels
 - Operational breakdown

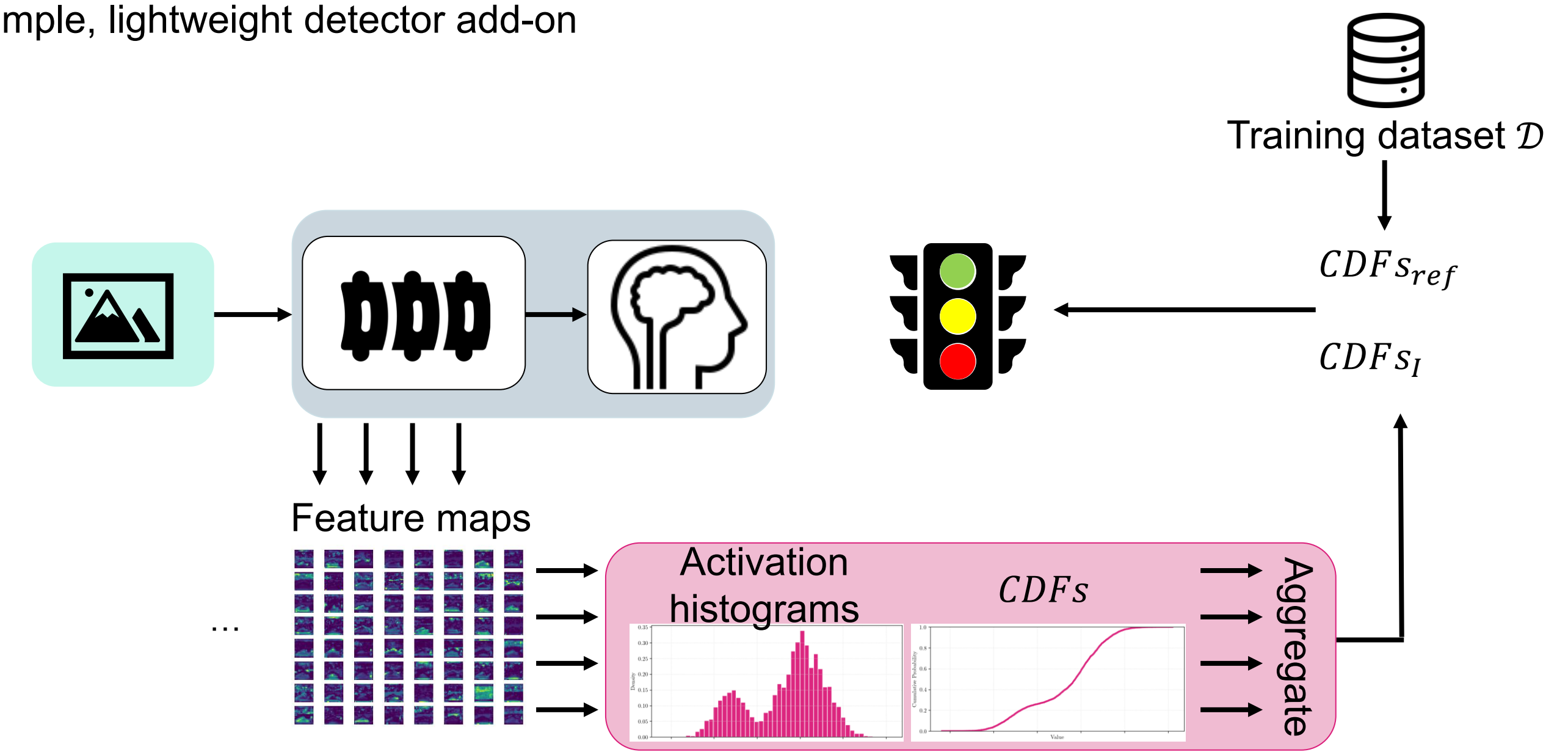


Baselines

- **Generative models**
 - Generative models learn the probability distribution of the in-distribution/training data
 - For OoD detection, an anomaly score is derived from it (e.g., negative log-likelihood)
- Density models
 - Flows (explicit, tractable likelihoods)
 - Variational autoencoders
- **Probabilistic detectors**
 - Variance networks using different scoring rules (Harakeh et al., 2021; [1])
 - Direct moment matching (DMM)
 - Negative log-likelihood (NLL)
 - Energy score (ES)
 - Top-k detection scores (Oksuz et al., 2022; [4])
 - Regression/localization
 - Mean squared error (MSE), NLL, entropy, ES, determinant, trace
 - Classification
 - Brier score, NLL, entropy, confidence score

Approach

- Activation-distribution monitor
- The frozen detector already encodes rich hierarchical information
- Summarize activations from selected backbone layers as channel histograms, convert them to cumulative distribution functions (CDFs)
- Compare test-image CDF_{S_I} to training-reference $CDF_{S_{ref}}$
- Simple, lightweight detector add-on



Evaluation

Model	Severity Levels AUROC (%) ↑			
	3	5	8	10
Normalizing Flows (FasterRCNN)				
NF head	53.55 ± 1.12	63.12 ± 1.17	71.34 ± 1.26	74.63 ± 1.31
NF backbone (C4, C5)	51.12 ± 1.15	61.53 ± 1.21	71.21 ± 1.21	76.53 ± 1.24
NF backbone (C1, C2, C3)	55.12 ± 1.21	73.56 ± 1.02	75.51 ± 1.16	76.61 ± 1.21
NF backbone	56.12 ± 1.16	73.45 ± 0.99	76.25 ± 1.14	77.72 ± 1.13
NF backbone & head	56.43 ± 1.24	73.32 ± 1.15	76.87 ± 1.22	77.89 ± 1.21
M-NF head	54.76 ± 1.31	64.41 ± 1.05	72.11 ± 1.20	74.93 ± 1.13
M-NF backbone (C4, C5)	55.76 ± 1.22	67.54 ± 1.18	73.71 ± 1.11	76.73 ± 1.12
M-NF backbone (C1, C2, C3)	58.61 ± 1.30	68.94 ± 1.11	75.81 ± 1.12	77.73 ± 1.11
M-NF backbone	58.76 ± 1.24	69.12 ± 1.01	73.91 ± 1.09	78.73 ± 1.07
M-NF backbone & head	61.75 ± 1.26	76.54 ± 1.34	77.10 ± 1.15	79.28 ± 1.27
Top-3 Detection Scores (FasterRCNN ES)				
Trace	52.32 ± 1.27	60.63 ± 1.09	64.84 ± 1.11	68.79 ± 1.00
Determinant	46.47 ± 1.31	47.94 ± 1.11	53.23 ± 1.17	57.16 ± 0.94
Entropy (Regression)	56.70 ± 1.15	60.72 ± 1.12	63.01 ± 1.21	68.72 ± 0.97
Confidence Score	58.83 ± 1.28	72.61 ± 1.09	75.56 ± 1.20	78.30 ± 1.01
Entropy (Classification)	62.84 ± 1.11	77.60 ± 1.10	81.21 ± 1.07	81.70 ± 1.03
Activation-Distribution (Ours) (FasterRCNN)				
CDFs head	49.02 ± 1.35	48.26 ± 1.31	56.22 ± 0.87	69.19 ± 1.04
CDFs backbone (C4, C5)	48.59 ± 1.29	62.17 ± 1.28	71.61 ± 1.01	78.03 ± 0.96
CDFs backbone (C1, C2, C3)	71.90 ± 1.33	79.46 ± 0.96	84.62 ± 0.85	86.73 ± 0.79
CDFs backbone	68.02 ± 1.31	80.26 ± 1.29	86.82 ± 1.16	87.39 ± 0.82
CDFs backbone & head	66.22 ± 1.35	77.66 ± 1.27	83.54 ± 1.11	86.52 ± 0.95

Detector	Model	Severity Levels AUROC (%) ↑			
		3	5	8	10
RT-DETR-I	CDFs backbone	76.40 ± 1.25	86.36 ± 0.79	91.38 ± 1.15	92.82 ± 1.12
RT-DETR-I	CDFs backbone & head	76.02 ± 1.22	86.22 ± 1.21	91.08 ± 1.31	92.21 ± 1.11
YOLOv9-m	CDFs backbone	77.56 ± 1.33	85.87 ± 0.93	90.56 ± 1.15	91.98 ± 1.22
YOLOv9-m	CDFs backbone & head	76.64 ± 1.32	84.87 ± 1.07	90.42 ± 1.21	90.81 ± 1.19
YOLOv10-m	CDFs backbone	77.75 ± 1.25	86.24 ± 0.79	90.87 ± 1.15	92.31 ± 1.12
YOLOv10-m	CDFs backbone & head	76.12 ± 1.22	86.12 ± 1.21	90.86 ± 1.31	91.23 ± 1.11
YOLOv11-m	CDFs backbone	78.07 ± 1.35	86.20 ± 0.94	90.51 ± 1.15	92.12 ± 1.03
YOLOv11-m	CDFs backbone & head	78.01 ± 1.29	84.58 ± 1.31	90.07 ± 1.28	92.01 ± 1.26

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Contributions

- Formulation
 - Define $P_{OR}(I) = 1 - P_{OOD}(I)$ as a practical data-side proxy for operational readiness
- Activation-distribution (CDFs)
 - Compare multi-layer channel activations with the training reference
- Systematic comparison
 - Benchmark against generative models and top-k detection scores
 - Detector-agnostic design

References

- [1] Harakeh et al.: “*Estimating and Evaluating Regression Predictive Uncertainty in Deep Object Detectors*”, ICLR, 2021
- [2] Hendrycks et al.: “*Benchmarking Neural Network Robustness to Common Corruptions and Perturbations*”, ICLR, 2019
- [3] Lin et al.: “*Microsoft COCO: Common Objects in Context*”, ECCV, 2014
- [4] Oksuz et al.: “*Towards Building Self-Aware Object Detectors via Reliable Uncertainty Quantification and Calibration*”, CVPR, 2022

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